5.1.2

(d) $\overline{x\overline{y} + y\overline{z} + \overline{x}z}$
$= (\overline{x\overline{y}})(\overline{y\overline{z}})(\overline{\overline{x}z})$
$= (\overline{x} + y)(\overline{y} + z)(x + \overline{z})$

5.3.2

(a) DNF

(b) Both

(c) CNF

(d) Neither

5.3.3

(d) $(x + y\overline{z})(\overline{x} + y + \overline{z})$
$= x\overline{x} + xy + x\overline{z} + \overline{x}y\overline{z} + y\overline{z} + y\overline{z}$ (distributive)
$= xy + x\overline{z} + \overline{x}y\overline{z} + y\overline{z}$ (complement, idempotent)
$= xy + x\overline{z} + y\overline{z}$ (distributive, annulment)

5.4.1

(b) $f = \overline{x}\overline{y}z + xy\overline{z}$ (minterms $m_1 + m_6$)
Only multiplication and complement: $f = \overline{\overline{\overline{x}\overline{y}z} \cdot \overline{xy\overline{z}}}$
Only addition and complement: $f = \overline{x + y + \overline{z} + \overline{x} + \overline{y} + z}$

5.4.2

(a) $f = x\overline{y} + \overline{x}y$ (XOR, minterms $m_1 + m_2$)
Only NAND: $f = \overline{\overline{x \cdot \overline{xy}} \cdot \overline{y \cdot \overline{xy}}}$
Only NOR: $f = \overline{\overline{x + y} + \overline{\overline{x + x} + \overline{y + y}}}$

5.5.1

(a) **Satisfied** when $x = True$, $y = False$, and $z = False$

(b) **Satisfied** when $x = True$, $y = False$, and $z = False$

(c) **Not satisfiable.**

x	y	z	$f(x, y, z)$
False	False	False	False
False	False	True	False
False	True	False	False
False	True	True	False
True	False	False	False
True	False	True	False
True	True	False	False
True	True	True	False

(d) **Satisfied** when $x = True$, $y = False$, and $z = False$

5.6.4

(b) $f = xy + xz + yz$

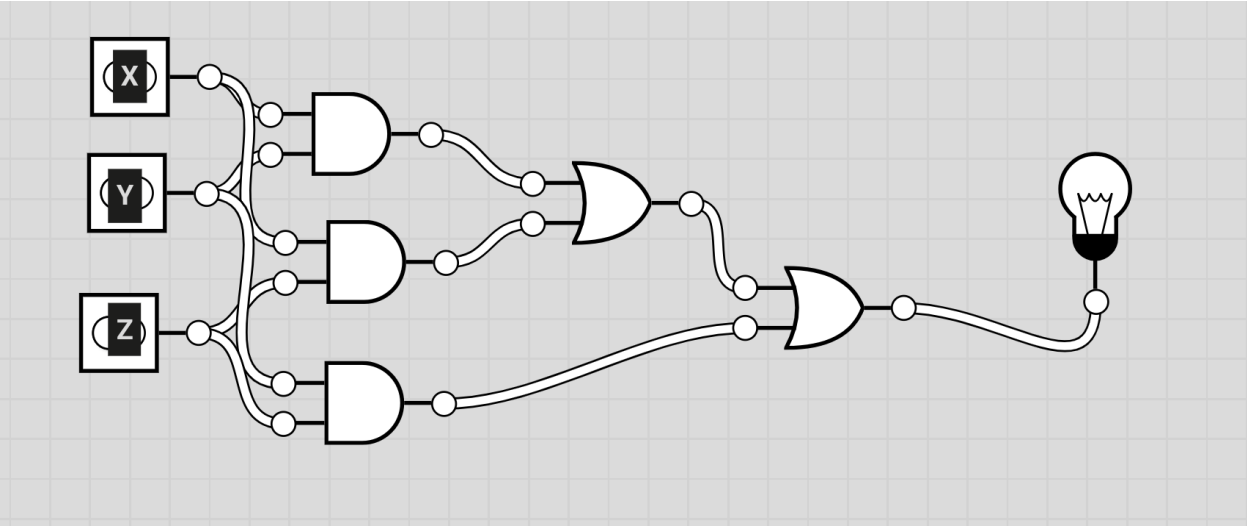


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5.4.5

(a) We will construct {+, complement} which is functional complete.
Since the complement operation is in both sets, it is sufficient to show that + can be computed using complement and $\rightarrow$. One of the conditional identities from the laws of propositional logic says that $x \rightarrow y$ is equivalent to $\overline{x} + y$. Therefore, $x + y$ is equivalent to $\overline{x} \rightarrow y$. Therefore, {complement, $\rightarrow$} is functional complete.

5.5.5

(a) $x_{A1} + x_{A2} + x_{A3}$ (1 iff A is in at least one period)

(b) $\overline{x_{A1}x_{A2} + x_{A1}x_{A3} + x_{A2}x_{A3}}$ (1 iff A is in at most one period)

(c) $\overline{x_{A1}x_{B1}} \cdot \overline{x_{A2}x_{B2}} \cdot \overline{x_{A3}x_{B3}}$ (1 iff A and B not in same period)

(d) $4n + 3m$ terms: n at-least-one clauses (1 per class) + $3n$ at-most-one clauses (3 pairwise clauses per class) + $3m$ incompatibility clauses (3 periods per forbidden pair)

5.6.3

(d) List the the law you apply. Please don't forget to draw the simplified circuit

$F(x, y) = \overline{\overline{x} \cdot \overline{y}} + \overline{y + \overline{y}}$
$= \overline{\overline{x} \cdot \overline{y}} + \overline{1}$ (Complement Law)
$= \overline{\overline{x} \cdot \overline{y}} + 0$ (Substitution)
$= (\overline{\overline{x}} + \overline{\overline{y}}) + 0$ (De Morgan's Law)
$= (x + y) + 0$ (Double Negation Law)
$= x + y$ (Identity Law)

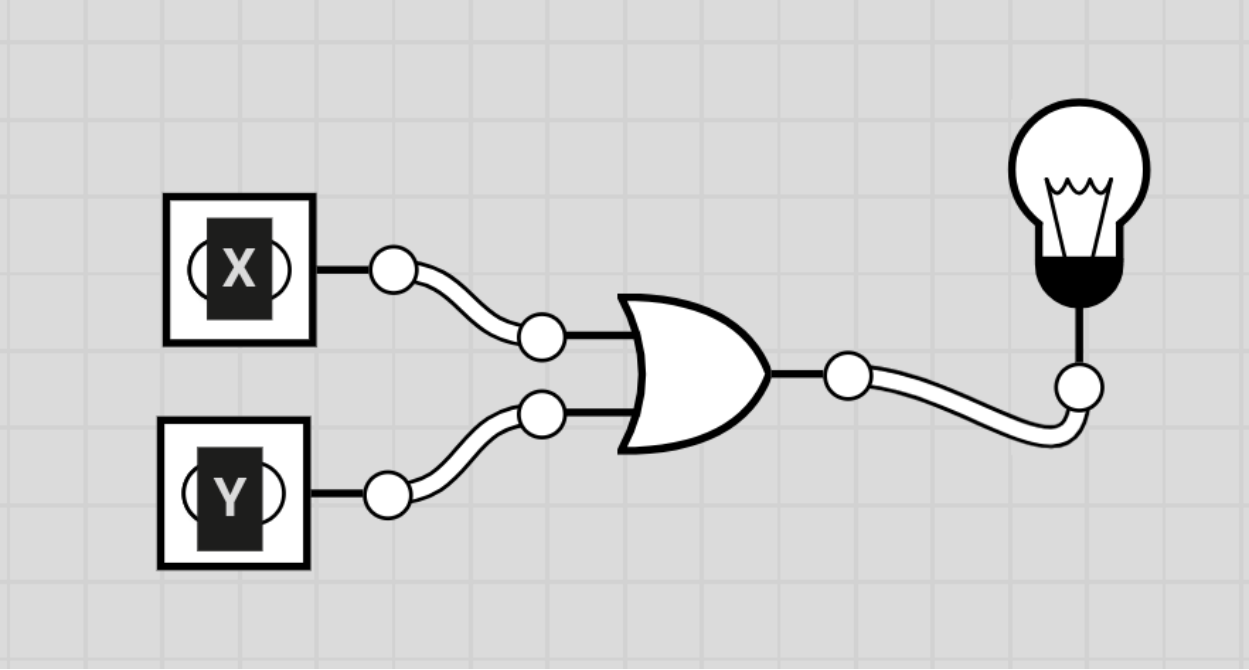


Figure 2: Pasted image 20261101171947.png